

TEST 5: COMPARISON TEST

Lesson 30

Given a series $\sum_{n=1}^{\infty} U_n$ where U_n -are positive terms

Comparison test state as follows:

* If $\sum_{n=1}^{\infty} U_n \leq \sum_{n=1}^{\infty} b_n$ and $\sum_{n=1}^{\infty} b_n$ is Convergent then $\sum_{n=1}^{\infty} U_n$ is also Convergent.

** If $\sum_{n=1}^{\infty} b_n \leq \sum_{n=1}^{\infty} U_n$ and $\sum_{n=1}^{\infty} b_n$ is divergent then $\sum_{n=1}^{\infty} U_n$ is also divergent.

STANDARD REFERENCE SERIES FOR COMPARISON

① The P-Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \text{Converges if } p > 1 \\ \text{diverges if } p \leq 1 \end{cases}$$

② GP Series

$$\sum_{n=1}^{\infty} ar^n = \begin{cases} \text{Converges if } |r| < 1 \\ \text{diverges if } |r| \geq 1 \end{cases}$$

The p-series and GP series are among standard reference series for $\sum_{n=1}^{\infty} b_n$ however other series can be used instead.

Example 1

Test whether the following series are Convergent/divergent?

$$(a) 1 + \frac{1}{2^2} + \frac{1}{3^3} + \frac{1}{4^4} + \frac{1}{5^5} + \dots + \frac{1}{n^n} + \dots$$

$$(b) \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} + \dots$$

$$(c) 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots + \frac{1}{n} + \dots$$

Solution

$$(a) \text{ We know: } 1 + \frac{1}{2^2} + \frac{1}{3^3} + \frac{1}{4^4} + \dots = \sum_{n=1}^{\infty} \frac{1}{n^n}$$

$$\leq 1 + \frac{1}{2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots = \sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$$

but $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$ is a GP-series with $r = \frac{1}{2} < 1$ and is Convergent

Hence by Comparison test $\sum_{n=1}^{\infty} \frac{1}{n^n}$ is Convergent.

$$(b) \text{ We know: } \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \dots$$

$$\leq \frac{1}{1 \times 1} + \frac{1}{2 \times 2} + \frac{1}{3 \times 3} + \frac{1}{4 \times 4} + \dots = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

but $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is a P-series with $p = 2 > 1$ and Convergent

Hence the given Series is Convergent by Comparison test.

(3)

(c) We know: $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \dots = \sum_{n=1}^{\infty} \frac{1}{n}$

$$> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \left(\frac{1}{16} + \dots\right)$$

$$= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$$

consider Sum of n terms:

ie $S_n = 1 + \frac{n}{2}, n \rightarrow \infty$

$$= \lim_{n \rightarrow \infty} \left(1 + \frac{n}{2}\right) = \infty \leftarrow \text{divergent}$$

Hence the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent by Comparison test.